Chapter 8

Campbell Biology, AP Edition, 12e

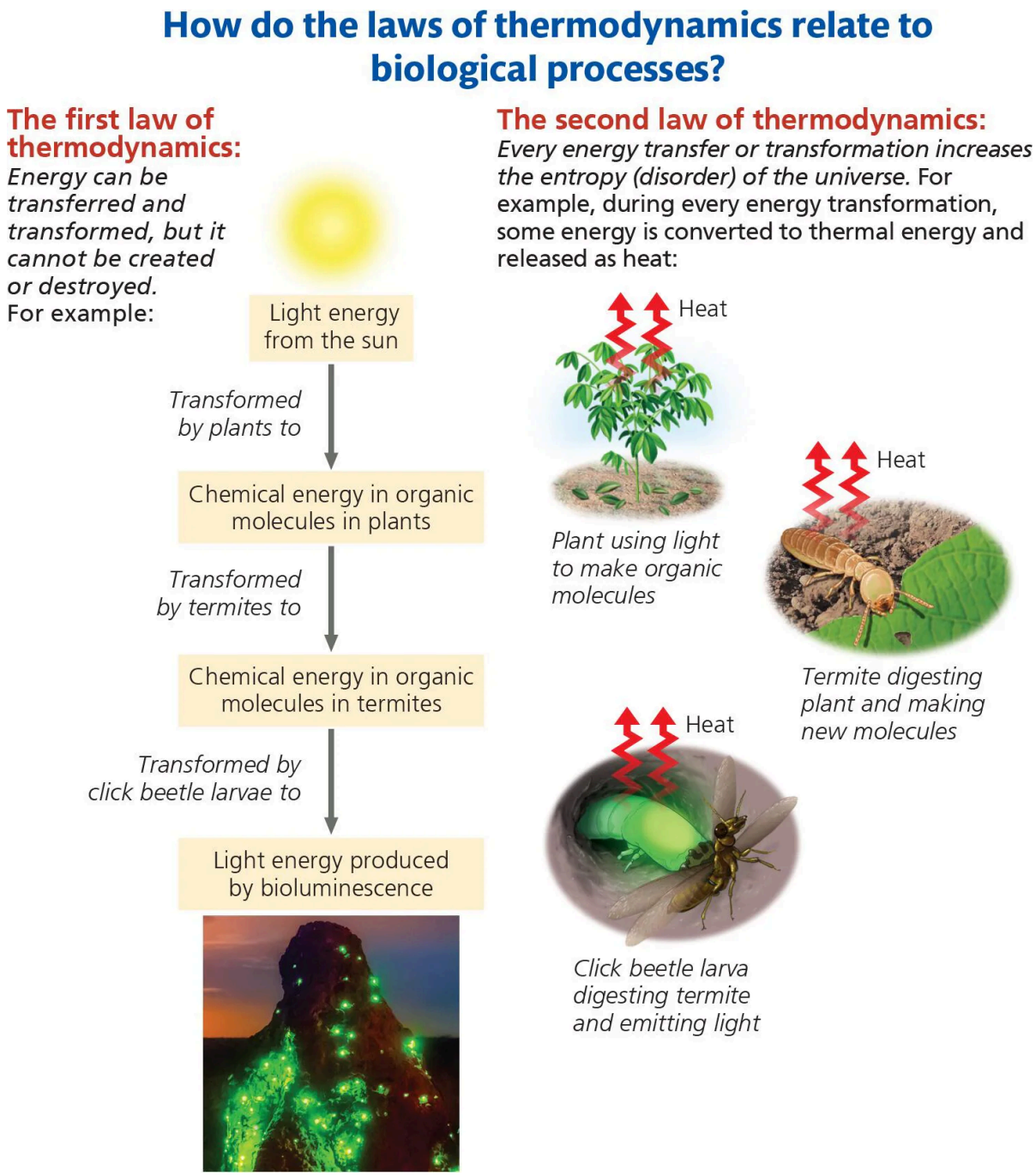
An image depicting a phenomenon known as bioluminescence, where organisms convert chemical energy to light energy. Pyrophorus nyctophanus, the click beetle, uses bioluminescence in its larval stages to attract termites from the termite mound for its food.

Chapter 8

An Introduction to Metabolism

Figure 8.1

The green glowing spots on the outside of this Brazilian termite mound are larvae of the click beetle, *Pyrophorus nyctophanus*. These larvae convert the energy stored in organic molecules to light, a process called bioluminescence, which attracts termites that the larvae eat. Bioluminescence and other metabolic activities in a cell are energy transformations that are subject to physical laws.



The first law of thermodynamics: Energy can be transferred and transformed, but it cannot be created or destroyed. For example: Light from the sun is transformed into chemical energy in organic molecules in plants. That is transformed by termites to chemical energy in organic molecules in termites. That is then transformed by click beetle larvae to light energy produced by bioluminescence.

The second law of thermodynamics: Every energy transfer or transformation increases the entropy (disorder) of the universe. For example, during every energy transformation, some energy is converted to thermal energy and released as heat. A plant using light to make organic molecules, or a termite digesting

plants and making new molecules, or a click beetle larva digesting termites and emitting light, all release heat.

Key Concepts

- **8.1An organism's metabolism transforms matter and energy****
- **8.2The free-energy change of a reaction tells us whether or not the reaction occurs spontaneously****
- **8.3ATP powers cellular work by coupling exergonic reactions to endergonic reactions****
- **8.4Enzymes speed up metabolic reactions by lowering energy barriers****
- **8.5Regulation of enzyme activity helps control metabolism****

All living organisms require a constant input of energy, which is used for organization, growth, and reproduction **(BIG IDEA 2)**. Common to life are core processes that depend on enzymes to regulate the chemical reactions that drive the metabolic activities of life **(BIG IDEAS 1 & 4).**

Study Tip

Make a table: Fill in the following table for each process you read about in this chapter, such as water spilling over a dam or a chemical reaction.

Process	Starting Materials; Relative Energy Level	Ending Materials; Relative Energy Level	Will this process occur spontaneously (without an input of energy)?
Water spilling over a dam	Water at the top of a dam; higher energy level	Water at the bottom of a dam; lower energy level	Yes
Hydrolysis of ATP (Figure 8.9)			

The table is as follows:

Process	**Starting materials; relative energy level**	**Ending materials; relative energy level**	**Will this process occur spontaneously (without an input of energy)?**
Water spilling over a dam	Water at the top of a dam; higher energy level	Water at the bottom of a dam; lower energy level	Yes
Hydrolysis of A T P (Figure 8.9)	blank	blank	blank

Get Ready for This Chapter

In this chapter, you will learn about the laws of thermodynamics (energy relationships) that govern metabolism, the chemical reactions that transform matter and energy in living organisms. Before beginning this chapter, you should be able to do the following:

- Explain the basic concepts of potential energy and kinetic energy (see Concept 2.2, section entitled “The Energy Levels of Electrons,” and Concept 3.2, section entitled “Temperature and Heat”).
- Describe what a chemical reaction is, and give an example (see Concept 2.4).
- Explain what chemical equilibrium is (see Concept 2.4).
- Describe how the shape of a protein is determined and how shape influences function (see Concept 5.4, section entitled “Protein Structure and Function,” and Figure 5.18).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements is true regarding potential energy and kinetic energy?

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Water at the top of a dam has kinetic energy; water falling through a dam has
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That's incorrect. Try again.

Incorrect. Review the following material: Potential energy is the energy possessed by matter due to its location or structure. Water at the top of a dam has potential energy due to its altitude. Kinetic energy is the energy of motion, such as the energy of falling water. Energy can be used to do work.

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Water at the top of a dam has potential energy; water falling through a dam has kin
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That's correct.

Correct! If you had trouble with this question, review the following material: Potential energy is the energy possessed by matter due to its location or structure. Water at the top of a dam has potential energy due to its altitude. Kinetic energy is the energy of motion, such as the energy of falling water. Energy can be used to do work.

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Water gains kinetic energy as it falls through a dam; water uses potential energy a
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That's incorrect. Try again.

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2. Which of the following is a chemical reaction?

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The formation of a covalent bond between two amino acids.
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That's correct.

Correct! If you had trouble with this question, review the following material: Chemical reactions involve the making and breaking of chemical bonds, leading to changes in the composition of matter. Ice melting changes the state of water rather than the composition of water, and hydrogen bonds do not change the composition of the molecules involved. Chemical reactions cannot create or destroy atoms but can only rearrange (redistribute) the electrons among them.

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Changing a carbon atom to a nitrogen atom by radioactive decay.
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That's incorrect. Try again.

Incorrect. Review the following material: Chemical reactions involve the making and breaking of chemical bonds, leading to changes in the composition of matter. Ice melting changes the state of water rather than the composition of water, and hydrogen bonds do not change the composition of the molecules involved. Chemical reactions cannot create or destroy atoms but can only rearrange (redistribute) the electrons among them.

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Making a hydrogen bond between a water molecule and a sugar molecule.
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That's incorrect. Try again.

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The melting of ice.
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That's incorrect. Try again.

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3. Chemical equilibrium is reached when

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concentrations of reactants and products remain the same.
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That's incorrect. Try again.

Incorrect. Review the following material: Chemical equilibrium is the point at which the forward and reverse reactions of a chemical reaction offset one another exactly. Equilibrium does not mean that the reactants and products are equal in concentration, but only that their concentrations have stabilized at a particular ratio. Reactions are still going on in both directions, but with no net effect on the concentrations of reactants and products.

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concentrations of products are higher than the concentrations of reactants.
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That's incorrect. Try again.

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there are equal concentrations of reactants and products.
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That's incorrect. Try again.

Incorrect. Review the following material: Chemical equilibrium is the point at which the forward and reverse reactions of a chemical reaction offset one another exactly. Equilibrium does not mean that the reactants and products are equal in concentration, but only that their concentrations have stabilized at a particular ratio. Reactions are still going on in both directions, but with no net effect on the concentrations of reactants and products.

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the forward and reverse reactions occur at the same rate.
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That's correct.

Correct! If you had trouble with this question, review the following material: Chemical equilibrium is the point at which the forward and reverse reactions of a chemical reaction offset one another exactly. Equilibrium does not mean that the reactants and products are equal in concentration, but only that their concentrations have stabilized at a particular ratio. Reactions are still going on in both directions, but with no net effect on the concentrations of reactants and products.

4. Which of the following is the strongest evidence that protein structure and function are correlated?

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Enzymes tend to be globular in shape.
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That's incorrect. Try again.

Incorrect. Review the following material: A polypeptide chain of a given amino acid sequence folds into a three-dimensional shape determined by the interactions responsible for secondary and tertiary structure. A protein's specific structure endows each protein with a specific function. If the pH, salt concentration, temperature, or other aspects of its environment are altered, the weak chemical bonds and interactions within a protein may be destroyed, causing the protein to unravel and lose its native shape, a changed called denaturation. Because it is misshapen, the denatured protein is biologically inactive. The observation that a denatured protein, one without its normal shape, is nonfunctional supports the idea that the shape of the protein is crucial to its function.

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Proteins have four distinct levels of structure and many functions.
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That's incorrect. Try again.

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Denatured (unfolded) proteins do not function normally.
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That's correct.

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Proteins can readily form hydrogen bonds with water.
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That's incorrect. Try again.

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Learning Objectives

After studying this chapter, you should be able to do the following:

- 8.1Identify the first two laws of thermodynamics and explain how they relate to biological systems.
- 8.2Explain what is meant by change in free energy and relate it to the following: spontaneous reactions, capacity to do work, stability, and exergonic and endergonic reactions.
- 8.3Use examples to illustrate the coupling of ATP hydrolysis to endergonic reactions.
- 8.4Explain how enzymes speed up metabolic reactions.
- 8.5Describe how regulation of enzyme activity helps control metabolism.
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